

LDX-3300 — Ultra-Low-Noise, High PSRR Linear Regulator (LDO)

Synthetic test datasheet for KB ingestion pipeline validation

Parameter	Value	Condition
Output Voltage Noise	9.5 μV_{rms}	10Hz to 100kHz
PSRR	70 dB	$f = 10\text{kHz}$
Dropout Voltage	120 mV	at 200mA load
Output Current	up to 300 mA	continuous
Quiescent Current	55 μA	no load
Line Regulation	0.02 %/V	typical
Load Regulation	0.04 %	typical

1. General Description

The LDX-3300 is an ultra-low-noise linear regulator intended for powering noise-sensitive analog circuitry such as precision references, instrumentation amplifiers, and current source front ends. Its low dropout architecture allows efficient regulation from a single DC input rail while keeping output voltage noise below 10 microvolts RMS across the 10 Hz to 100 kHz band, which is critical for not degrading the noise floor of downstream zero-drift amplifier stages.

2. Generating Multiple Rails from a Single DC Input

For designs that must operate from a single DC supply but require a regulated positive rail, a regulated negative rail, and a clean voltage reference, a common architecture uses one LDX-3300 for the positive analog rail, a charge-pump-based or inverting switching regulator followed by a second LDX-3300 to generate a clean negative rail, and a dedicated low noise reference IC for the precision voltage reference. This approach isolates noisy switching elements from the precision signal path while still allowing fully single-supply system operation.

3. Decoupling and PCB Layout

Place the input and output decoupling capacitors as close as possible to the LDX-3300 pins, with a low ESR ceramic capacitor of at least 1 μF on the output to maintain loop stability and minimize output noise. Keep the ground return path for the regulator separate from sensitive analog ground traces where possible, using a star-ground or partitioned ground plane strategy on the PCB.